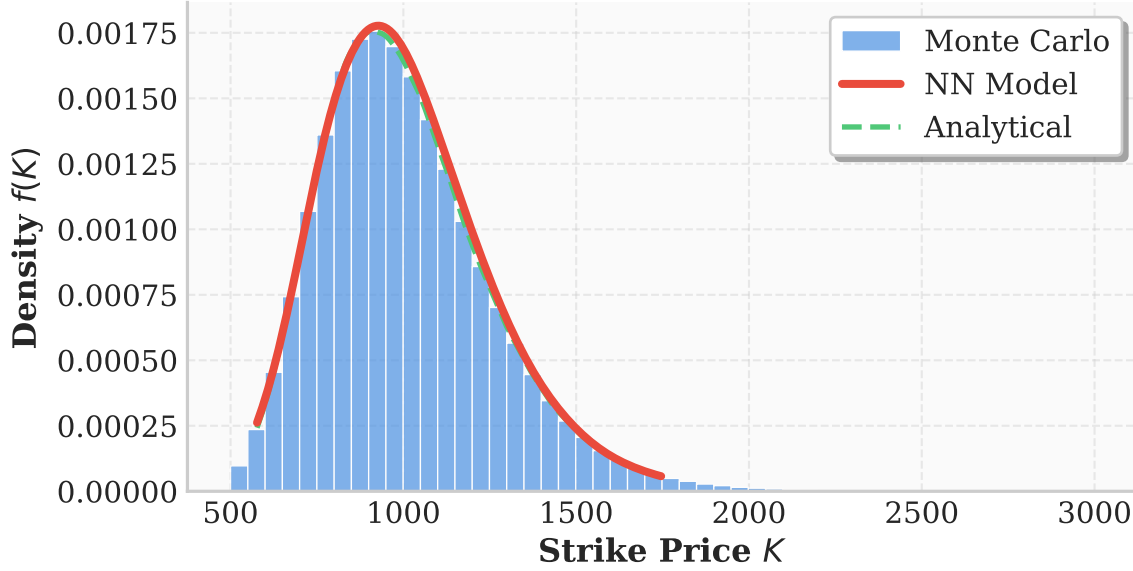


PDF Analysis: Neural Network Synthetic Local Volatility Model

Monte Carlo under ground-truth $\sigma: dS_t = rS_t dt + \sigma(t, S_t/S_0)S_t dW_t$

MC source: data

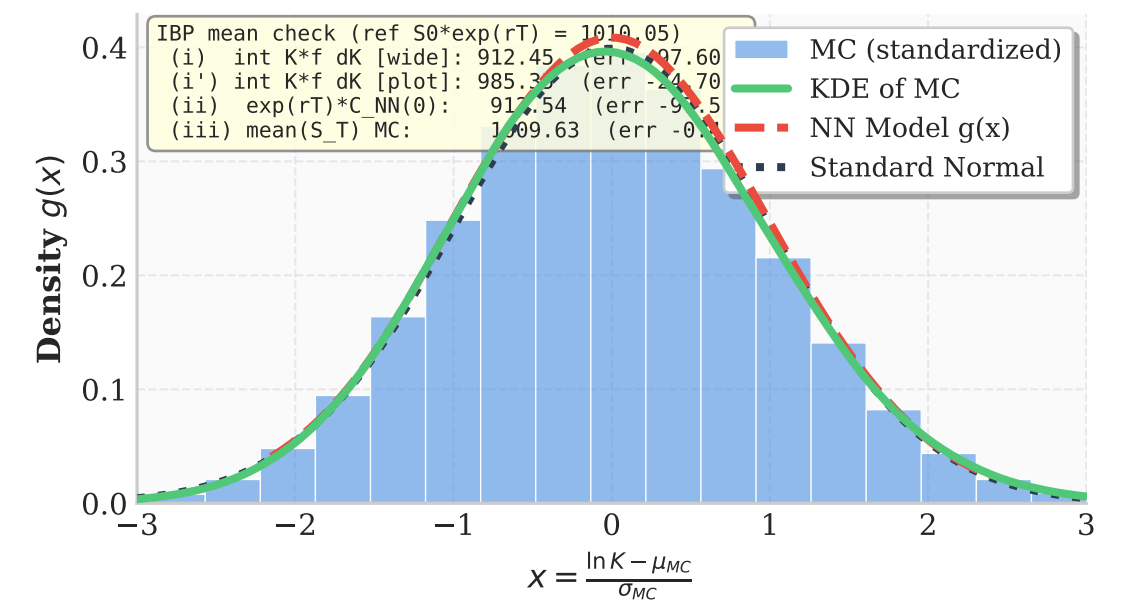
Strike Distribution (T = 0.25)



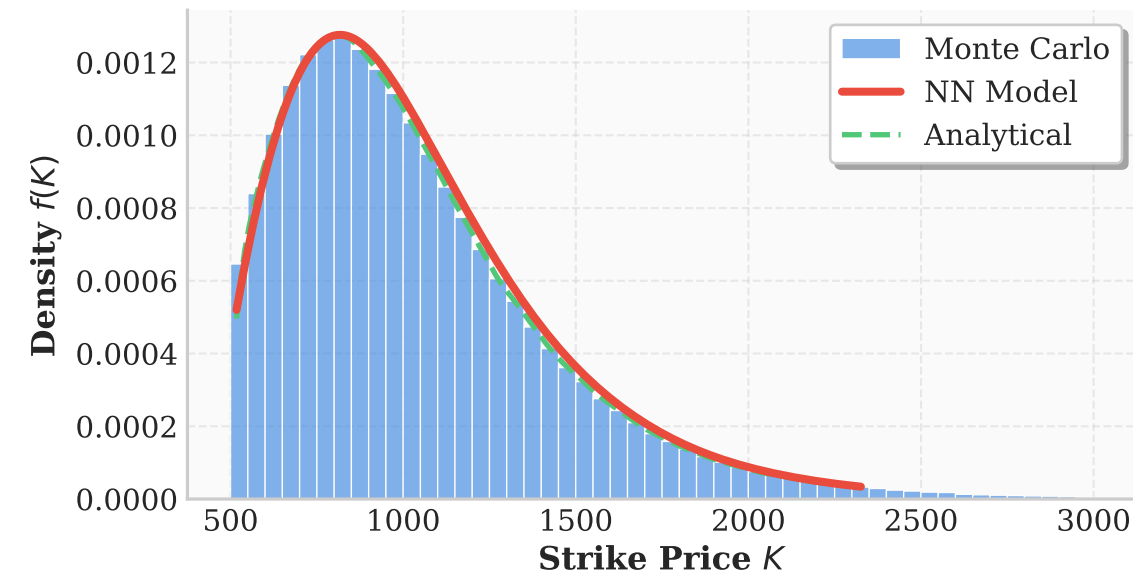
Log-Normal Distribution (T = 0.25)



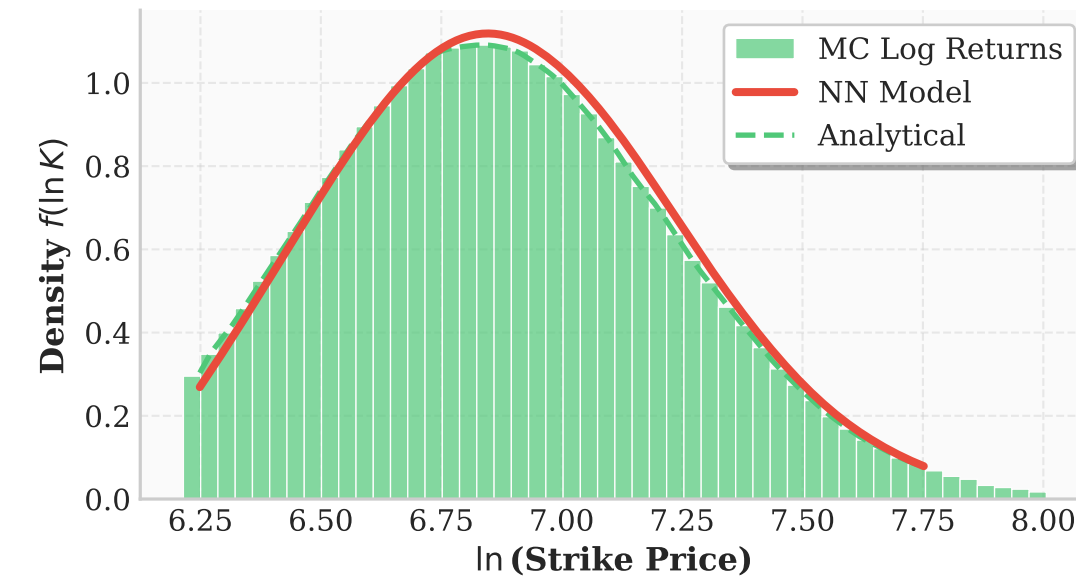
Gaussian Space (T = 0.25)
MC Standardization: $\mu_{MC}=6.89, \sigma_{MC}=0.24 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.89, \sigma_{model}=0.22$
MC Stats: $\mu=0.00, \sigma=1.00, \text{Skew}=0.09, \text{ExKurt}=0.03$



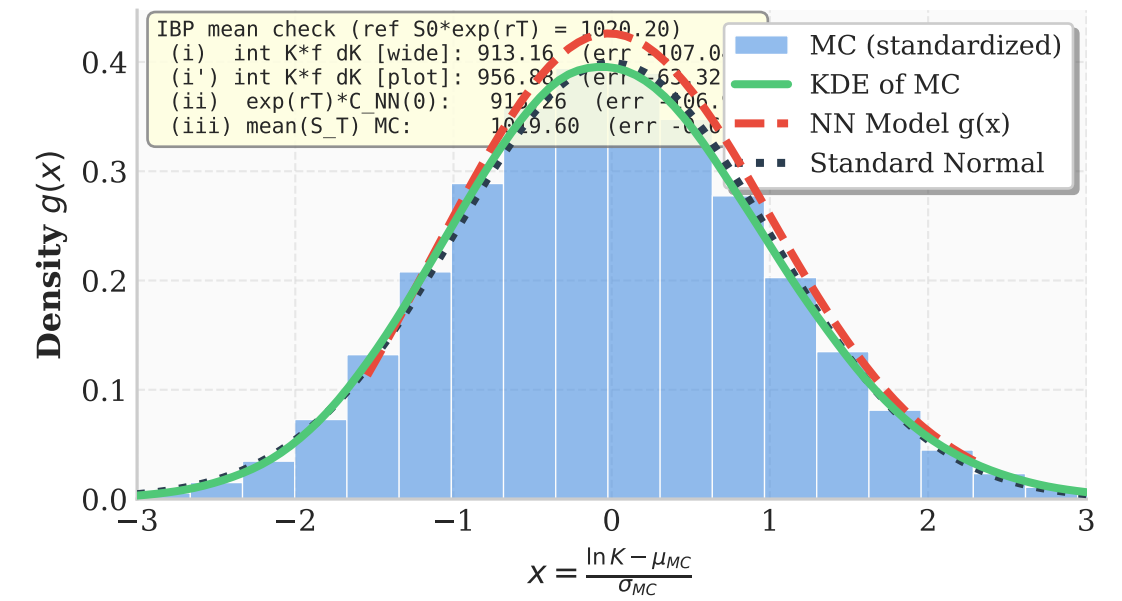
Strike Distribution (T = 0.50)



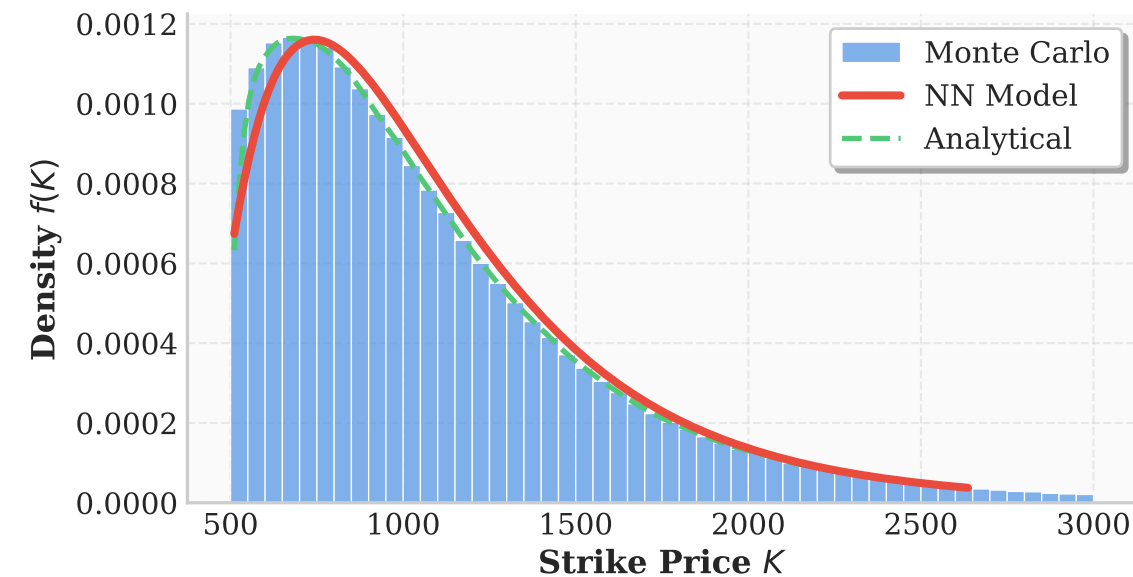
Log-Normal Distribution (T = 0.50)



Gaussian Space (T = 0.50)
MC Standardization: $\mu_{MC}=6.85, \sigma_{MC}=0.38 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.89, \sigma_{model}=0.33$
MC Stats: $\mu=0.00, \sigma=1.00, \text{Skew}=0.12, \text{ExKurt}=0.02$



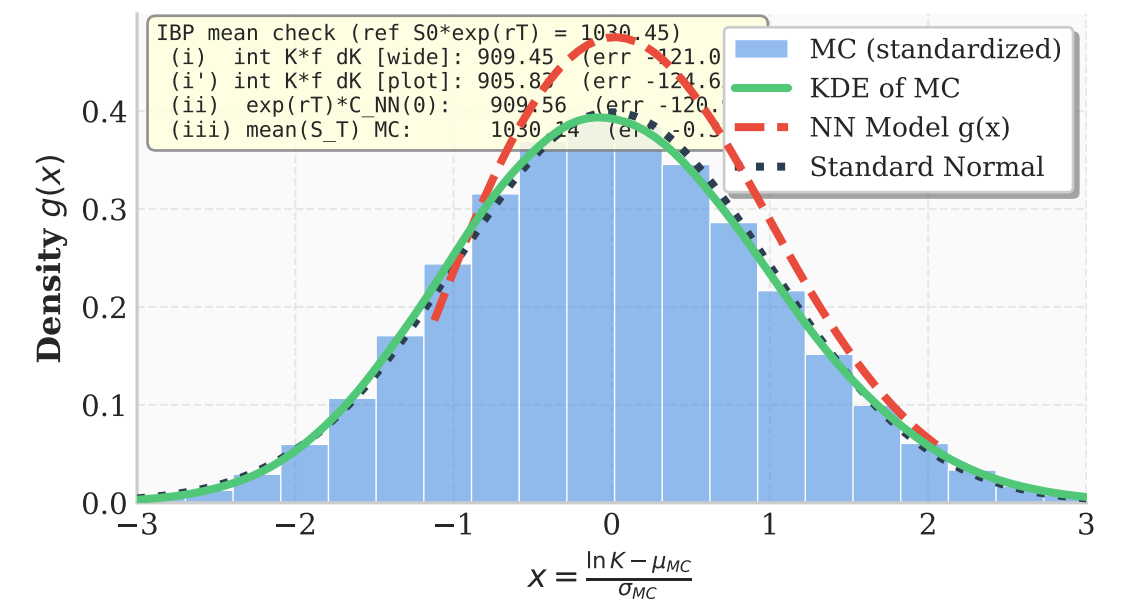
Strike Distribution (T = 0.75)



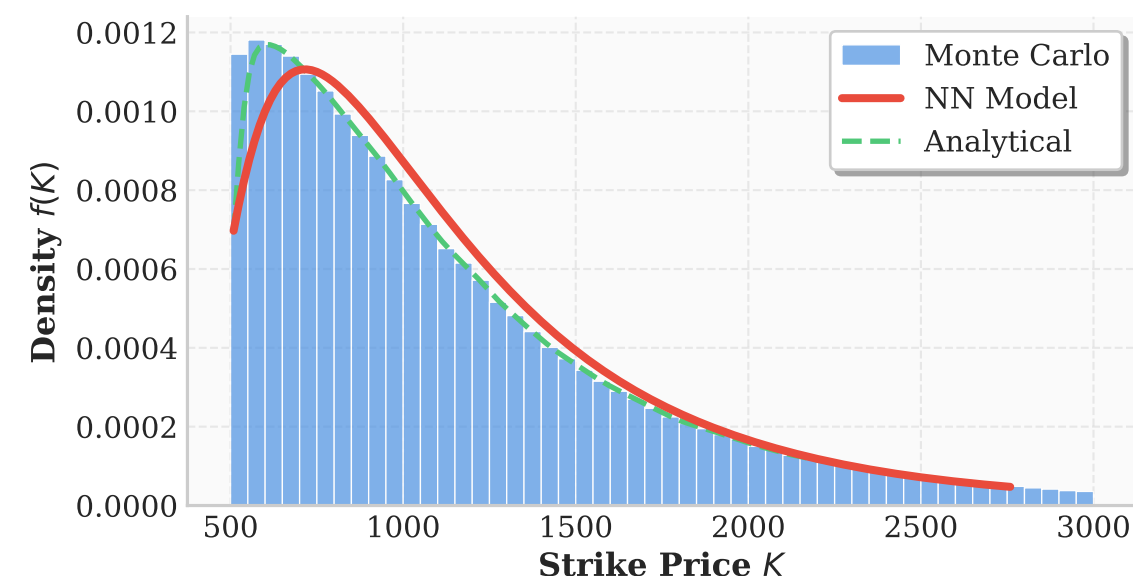
Log-Normal Distribution (T = 0.75)



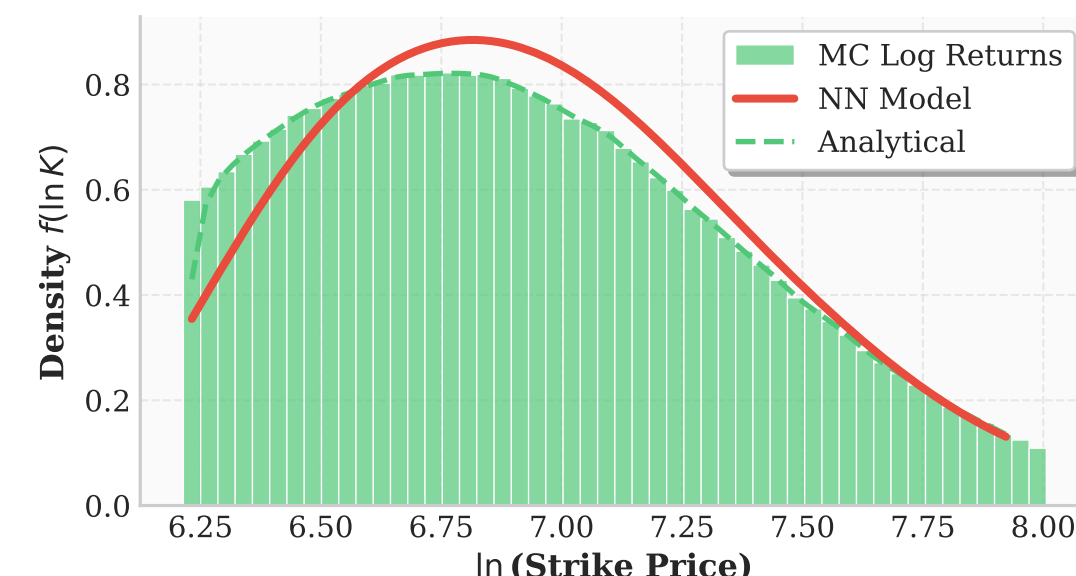
Gaussian Space (T = 0.75)
MC Standardization: $\mu_{MC}=6.81, \sigma_{MC}=0.50 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.92, \sigma_{model}=0.39$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=0.11, \text{ExKurt}=-0.04$



Strike Distribution (T = 1.00)



Log-Normal Distribution (T = 1.00)



Gaussian Space (T = 1.00)
MC Standardization: $\mu_{MC}=6.77, \sigma_{MC}=0.60 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.95, \sigma_{model}=0.41$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=0.07, \text{ExKurt}=-0.10$

